

Thermodynamic Chess

A chess engine as a statistical-mechanical system: theory, design decisions, and what an airtight implementation might measure

Conceptual summary of the `fablethermochess` engine · July 2026

ABSTRACT. Conventional engines value a chess position by minimax: the value of a position is the value of its single best continuation. This engine replaces that rule with equilibrium statistical mechanics. The legal moves of a position are treated as the microstates of a canonical ensemble; each carries a Boltzmann weight; and the value of the position is its *free energy* $F = T \ln Z$, which decomposes into an energetic part (how good the moves are) plus an entropic part (how many good moves there are). The temperature is not a tuning knob: it is solved per-position from a fixed-heat-capacity constraint $\text{Var}(Q)/T^2 = C$, which turns out to define a genuine phase structure — positions where the demanded capacity is achievable, positions that pin at their own critical capacity, and frozen (forced) positions. Every component of the engine — piece values, quiescence, pruning, mate handling, draw handling, and the per-piece "chemical potential" — is derived from, or reconciled with, this single physical picture. The engine plays the argmax move; the thermodynamic state variables (T , S , F , C^* , μ) are measurements taken along the way, with Stockfish attached as an external reference thermometer.

1. The central idea

A chess position with the player to move is treated as a physical system in contact with a heat bath. Its accessible microstates are its legal moves. Each move a has a value Q_a — the (negated, see below) energy of the microstate — obtained recursively from the position it leads to. The system occupies its microstates with canonical (Boltzmann) probabilities:

$$\pi_a = e^{Q_a/T} / Z, \quad Z = \sum_b e^{Q_b/T} \quad (1)$$

We adopt the convention $\mathbf{Q} = -\mathbf{E}$: good moves are low-energy states, and maximizing Q is minimizing energy. Under this convention the free energy of the position (with $k_B = 1$) is

$$F = T \ln Z = \langle Q \rangle + T \cdot S, \quad S = -\sum_a \pi_a \ln \pi_a \quad (2)$$

and the engine *maximizes* F , which corresponds to the usual thermodynamic statement that equilibrium minimizes $E - TS$. The decomposition in (2) is the heart of the design: a position is valuable both for the quality of its best continuations ($\langle Q \rangle$, weighted toward the best by the Boltzmann factor) and for its *optionality* — the entropy S of having many comparably good continuations, paid at the going temperature T .

2. Free energy replaces minimax

Search proceeds in negamax form. The value of a move is minus the free energy of the resulting position for the opponent, and the value of a position is the log-sum-exp ("softmax") of its move values:

$$Q_a = -F(\text{position after } a), \quad F = \max_a Q_a + T \ln \sum_a e^{(Q_a - \max Q)/T} \quad (3)$$

Both sides are computed by the identical rule — equilibrium requires this symmetry; there is no separate "opponent model." Two limits anchor the construction. As $T \rightarrow 0$, $F \rightarrow \max_a Q_a$: the search becomes exact

minimax, and the engine is a conventional engine. As $T \rightarrow \infty$, π becomes uniform and $F \rightarrow \langle Q \rangle + T \ln n$: pure optionality. Real play lives between the limits, and *where* between them is not chosen by hand but solved for (§4).

Connections. Equation (3) is exactly the *soft Bellman backup* of maximum-entropy reinforcement learning (soft Q-learning), and a two-player game in which both sides best-respond with Boltzmann distributions is a *logit quantal-response equilibrium*. What distinguishes this engine from both is that T is not a hyperparameter: it is an emergent, position-dependent quantity fixed by a thermodynamic constraint. The engine can be read as a self-consistent-temperature QRE player.

The engine ultimately *plays* $\arg\max_a Q_a$ — it does not sample from π . Temperature enters play indirectly: through the softmax backups of interior nodes (which reward optionality below the root) and through the pruning width (§8). The weights π_a are displayed as a measurement of how contested the choice was.

3. The energy function: geometry, not convention

Leaf positions need a static energy. The design rule is that *nothing is imported from chess folklore* — no 1/3/3/5/9 point table, no hand-tuned piece-square bonuses. Everything derives from board geometry:

- **X-ray mobility** $x(t, sq)$: the number of squares a piece of type t sweeps from square sq on an *empty* board — a pure function of geometry (rook: 14 everywhere; bishop: 7–13; knight: 2–8; queen: 21–27).
- **Material base**: each piece type's base value is its mean x-ray mobility over the 64 squares, $base(t) = \langle x(t, \cdot) \rangle$. The base is what a piece is worth for existing; it is the same quantity as mobility, averaged, which makes the material term dominate the positional term without introducing a second unit.
- **Real mobility**: at quiet leaves the engine adds each piece's blocker-aware mobility — squares actually reachable, rays stopped at the first blocker — so open files, blocked bishops, and centralization are seen, not assumed.

The static energy of a position for the side to move is then

$$U = \sum_{\text{own pieces}} [base(t) + mob(t, sq)] - \sum_{\text{their pieces}} [base(t) + mob(t, sq)] \quad (4)$$

with two principled exceptions. The **king** contributes nothing: it is never traded, both sides always have exactly one (so any base cancels), and rewarding its mobility would teach the engine to walk its king into the open with no compensating safety term. The **pawn** contributes a fixed base of 2 (its typical move count) and no mobility term: pawn mobility is dominated by enemy-dependent capture geometry, and rewarding pawn x-ray leads to systematic pathologies (a start-rank bonus that penalizes advancing). One pawn = 2 internal units defines the display unit ($\hat{\Delta}$).

Averaging (4) over squares gives the engine's *emergent exchange rates*, which are a falsifiable output of the theory rather than an input:

	Pawn	Knight	Bishop	Rook	Queen
mean x-ray mobility	(2)	5.25	8.75	14	22.75
emergent value ($\hat{\Delta}$)	1	5.25	8.75	14	22.75
classical value ($\hat{\Delta}$)	1	3	3	5	9

The geometric values compress pawns and spread the pieces: relative to tradition this engine treats pawns as cheaper, bishops as clearly better than knights (ratio 5:3), and heavy pieces as more precious. Systematic disagreements with Stockfish's evaluation (§10) are therefore not all errors — some are the measurable signature of valuing chess by geometry alone.

4. Emergent temperature and the phase structure

The one place a free parameter could hide is T . It is eliminated by prescribing, instead of a temperature, a **heat capacity** C and demanding the equilibrium fluctuation identity

$$C = \text{Var}_\pi(Q) / T^2 \quad (5)$$

hold at every node — the same identity that relates energy fluctuations to heat capacity in a canonical ensemble. Since π itself depends on T , (5) is a fixed-point equation. Writing $r(T) \equiv \sigma_\pi(T)/T$, the equation reads $r(T) = \sqrt{C}$, and the structure of r governs everything: $r \rightarrow 0$ as $T \rightarrow 0$ (the distribution freezes onto the best move, σ collapses exponentially) and $r \rightarrow 0$ as $T \rightarrow \infty$ (σ saturates at the spread of the move values while T grows), so r rises to a peak and falls. Each position therefore possesses an intrinsic **critical capacity**

$$C^* = \max_T \text{Var}_\pi(Q)/T^2 \quad (6)$$

— the largest heat capacity its move ensemble can sustain — and the solver distinguishes three phases:

1. **SUBCRITICAL** ($C < C^*$): the fixed point exists. Of the two crossings, the engine takes the one on the descending branch of r — the thermodynamically stable solution — located by geometric sweep and log-space bisection, solved to better than 1%. Here the dashboard shows $C_{\text{eff}} = C$ exactly.
2. **SUPERCRITICAL** ($C > C^*$): no temperature satisfies (5). The demand is unphysical for this position, and T pins at the **spinodal** $\hat{T} = \text{argmax } r(T)$, the closest achievable equilibrium; the achieved capacity saturates at $C_{\text{eff}} = C^* < C$. C thus acts as a capacity *demand* — an upper bound the position meets when it can.
3. **FROZEN** ($C^* \approx 0$): a single move dominates at every temperature — forced recaptures, mating sequences. T falls to a numerical floor (10^{-3} , far below the ~ 1 -unit evaluation resolution) and $F \rightarrow \max Q$: the minimax limit is recovered exactly where chess is genuinely forcing.

Robustness against absorbing states. The solver's search domain is set by a median-based (MAD) scale of the *bulk* of the Q distribution, never the full range. A mate score among quiet moves is an outlier of order 10^5 ; allowing it to define the temperature scale would open a spurious high-temperature branch in which F values the node as a lukewarm mixture of "mate" and "quiet move." Physically, terminal states are absorbing, not thermal (§6); numerically, the MAD domain ensures they freeze the ensemble instead of heating it. A near-critical subtlety is handled explicitly: when the band where $r \geq \sqrt{C}$ is narrower than one sweep step, a peak refinement detects the crossing and recovers the exact solution rather than misclassifying the position as supercritical.

4.1 The capacity schedule

C is scaled by the material remaining on the board: the value-weighted material fraction maps linearly from $C = 2.0$ (full board) down to $C = 0.6$ (bare board), times a user multiplier (0.3–1.5). Larger C demands lower T : the crowded opening runs *cool and decisive* — and is typically supercritical, pinned at its own $C^* \approx 1$ — while the emptying board relaxes the demand and lets the endgame run *hot and exploratory*, where the optionality term genuinely shapes play. Because pinning saturates gracefully, no setting of the slider can collapse the temperature

by accident; the old failure mode ("keep $C \leq 1.5$ or T dies") is eliminated, and C_{eff} on the dashboard reports which phase the position is in.

5. Local equilibrium: every node has its own temperature

Equation (5) is solved independently at every interior node, so temperature is a *local* quantity — the tree is in local thermal equilibrium rather than global. A sharp tactical node sits cold within an otherwise warm search; a maneuvering node sits warm within a forcing line. This is deliberate: T measures the decision structure of a single position (how contested its move choice is at the demanded capacity), not a global property of the game. The graph of root temperature across a game is then a legitimate time series of a state variable, and its systematic drift — cool opening, warm middlegame, cooling endgame as C relaxes but spreads narrow — is one of the objects this engine exists to measure.

6. Absorbing states: mate and repetition

Two kinds of position are terminal and must not be treated thermally.

Checkmate is a ground state, not an energy level. A mated position returns -10^5 ; no entropy may attach to a terminal outcome. The backup rule therefore switches to pure minimax whenever the best move is a forced win (or every move a forced loss): $F = \max Q$ with no $T \ln Z$ term. Without this rule the softmax would award "many ways to mate" a free energy *above* the mate score itself — $F = \max Q + T \ln Z > \text{mate}$ — and the engine would prefer keeping mating options open to actually mating (an error the test suite now checks: during development it produced exactly this check-shuffling pathology). Mate scores also carry distance: every negation on the way up ticks the score one unit toward zero, so mate-in-2 strictly outranks mate-in-4, the engine converts won positions, and — because the scores stay node-relative — transposition-table entries remain valid across paths.

Repetition is an absorbing draw, $F = 0$. Any position recurring on the current search path scores zero, and the search is seeded with the hashes of every *played* position since the last irreversible move (capture or pawn push), so the engine also sees repetitions relative to the actual game: it avoids repetition when ahead and steers into it when lost. Zero is the natural draw value in this evaluation: U of (4) is antisymmetric between the sides, so 0 is exact material-mobility balance. Castling-rights changes need no special handling — rights are hashed, so superseded positions simply never match.

7. Quiescence as relaxation to local equilibrium

Thermodynamic state variables are only meaningful for systems at (local) equilibrium, and a position with hanging pieces is not one: it is mid-collision. The engine therefore separates timescales. Fast degrees of freedom — captures, promotions, check evasions — are relaxed *deterministically* (alpha-beta at $T = 0$, with MVV-LVA ordering and delta pruning) by a quiescence search until a quiet position is reached; only then is the static energy (4) evaluated and only quiet positions ever enter an ensemble. This mirrors the standard physical picture: collisions equilibrate fast modes long before the slow modes are sampled thermally. Inside quiescence the king ranks as the *last* attacker in capture ordering — its ordering value is its irreplaceability, not its (zero) exchange value.

8. Partition-function truncation: the softmax analog of alpha-beta

Alpha-beta cutoffs are unsound here — every child contributes to Z — but contributions decay exponentially. At depth ≥ 2 the engine runs a shallow ranking pass (depth -2) over all children, solves (5) on the shallow values to get a provisional T , and fully recurses only children within $3T$ of the best (capped at 16); the tail keeps its shallow values, contributing its ($<1\%$) share of Z at reduced accuracy rather than being discarded. The pruning width is thus *temperature-scaled*: cold, forcing nodes truncate to almost a single move — recovering alpha-beta-like efficiency exactly where minimax is exact — while hot nodes keep many live branches. Pruning aggressiveness is not a policy choice; it is read off the node's own thermodynamic state.

9. Optionality at the leaves; parity symmetry

Interior nodes get optionality for free — it is the $T \cdot S$ term of (2), propagated by the backup. Leaves need a static proxy, and the engine offers two, selected by `OPTIONALITY_MODE`:

- **'xray'** (default): the real-mobility term of (4) stands in for leaf optionality. Cheap, side-symmetric, and in the same units as everything else.
- **'thermo'**: the leaf's own moves are treated as microstates, and optionality is $T \cdot S$ of the Boltzmann distribution over their static values — the same thermodynamics as the rest of the engine, evaluated at the leaf. This resolves the conceptual wrinkle that a single position "has no ensemble": it does — the moves available from it.

The thermo mode carries a symmetry obligation. Scoring only the mover's optionality would hand whoever happens to be on move at the leaf an entropy bonus, flipping sign with leaf parity and injecting a systematic odd/even-depth bias. The implementation therefore evaluates *both* ensembles — the opponent's via a null move (a pass: the exact formalization of "what could they do from here") — and scores the difference, $TS_{us} - TS_{them}$. In check there is no equilibrium optionality (passing is illegal; the ensemble is not stationary) and the term vanishes. In the mirror-symmetric starting position the symmetric optionality is zero to machine precision, which the test suite asserts.

10. Chemical potential: which pieces carry the position

Let $p_P = \sum_{a \text{ from } P} \pi_a$ be the Boltzmann weight of all moves originating from piece P . Removing P 's moves from the partition function (holding the others fixed — an ideal-gas/independent-pieces approximation) changes the free energy by

$$\mu_P = F - F_{-P} = -T \ln(1 - p_P) \quad (7)$$

the free-energy cost of freezing P in place: a per-piece activity measure in the same units as F . A piece with $\mu_P \approx 0$ is thermodynamically idle regardless of its material worth; a piece carrying most of the ensemble weight is, in a precise sense, the piece the position is *about*. The dashboard renders μ for every mobile piece of the side to move, sorted.

11. The measurement apparatus

The engine is also an instrument, and its panels are its gauges: the state grid (U_w , U_b , T , S , $\langle Q \rangle$, TS , F , C demand vs C_{eff} with phase annotation), the move-ensemble table (Q_a , π_a), the chemical potentials, and per-move time series of Eval , T , and S . Two conventions keep the numbers honest. First, the headline **Eval** is everywhere the value of the move actually played ($\text{argmax } Q$, in White-POV pawns) — the energetic part alone — while F , which adds the optionality term, is displayed separately; comparing F against a minimax engine would conflate the two. Second, **Stockfish never influences play**: it runs in its own worker as an external reference thermometer, and the $\Delta(\text{engine-SF})$ column is a running measurement of where geometric-thermodynamic valuation diverges from conventional valuation — part noise, part the emergent exchange rates of §3, and occasionally the optionality-blindness of minimax itself.

12. What an airtight implementation might tell us

- **$C^*(\text{position})$ is a new invariant.** The critical capacity (6) is a parameter-free property of a position's move ensemble — how much evaluation fluctuation it can sustain. Its trajectory over master games (does it peak in complex middlegames? collapse in technique phases?) and its correlation with human notions of "sharpness" and practical difficulty are directly measurable with this engine.
- **Phase geography of a game.** Openings here are typically pinned (supercritical), endgames subcritical, mates frozen. Where the transitions fall — and whether strong play systematically steers toward or away from the pinned phase — is an empirical question the dashboard now answers per move.
- **Entropy as a playing style.** With the argmax convention fixed, the entire behavioral difference from minimax lives in the interior $T \cdot S$ terms. Games between this engine and its own $T \rightarrow 0$ limit isolate the value of optionality in chess — how many pawns a position of "many good moves" is actually worth.
- **Geometric value theory.** The $\Delta(\text{engine-SF})$ series quantifies how far pure board geometry lands from four centuries of empirical piece values — and in which position classes the gap closes.

Appendix: equation → implementation map

Object	Where it lives
Boltzmann weights π_a , S, F (eq. 1–3)	<code>computeF()</code> , <code>computeThermodynamics()</code>
Self-consistent T, phases (eq. 5–6)	<code>robustT()</code> , <code>_sigmaOverT()</code>
Static energy U (eq. 4), geometric bases	<code>fast_real_u()</code> , <code>xrayMobility()</code> , <code>PIECE_BASE</code>
Capacity schedule C(material)	<code>materialC()</code> , <code>C_OPEN/C_BARE</code>
Absorbing mates, distance ticking	<code>tickMate()</code> , mate guard in <code>computeF()</code>
Repetition seeding from the played game	<code>collectPastKeys()</code> → <code>PAST_KEYS</code>
Quiescence relaxation (§7)	<code>quiesce()</code> , <code>staticEval()</code>
Partition-function truncation (§8)	<code>thermoSearch()</code> , <code>BOLTZ_SIGMA</code> , <code>TRUNC_MAX</code>
Symmetric leaf optionality (§9)	<code>leafOptionality()</code> , <code>fast_make_null()</code>
Chemical potential μ_p (eq. 7)	<code>computeChemicalPotentials()</code> , μ panel
Reference thermometer	<code>initStockfish()</code> et seq. (never influences play)